U1
MCP23017-E/SP / 0x20
PANEL_3V3
SCL
SDA
SCK
SDA
GPA0
GPA1
GPA2
GPA3
GPA4
GPA5
GPA6
GPA7
INTA
INTB
RESET
GPB0
GPB1
GPB2
GPB3
GPB4
GPB5
GPB6
GPB7
A0
A1
A2
VSS
VDD
BTN1
BTN2
BTN3
BTN4
BTN5
BTN6
BTN7
HOLD_IN
STOP1_IN
STOP2_IN
METER_EN_IN
IO_RESET
GND
J1
TO HAT J7 / I2C AND IRQ
SCL
SDA
HAT_TP_IRQ
HAT_ADC_CS
PANEL_3V3
GND
J5
TO HAT J6 / SPI
HAT_MISO
HAT_MOSI
HAT_SCLK
HAT_LCD_CS
PANEL_3V3
GND
J6
TO HAT J5 / CONTROL
HAT_LCD_DC
HAT_LCD_RST
HAT_LCD_BL
HAT_TP_CS
PANEL_3V3
GND
J7
5V IN / FROM HAT J12
PI_5V_IN
GND
J8
EXISTING 5V FAN / PASS
PI_5V_IN
GND
F1
MF-RHT050-2 / 500mA HOLD
PI_5V_IN
LCD_5V
J2
WAVESHARE 16446 / 2.8in TOUCH
LCD_5V
GND
LCD_MISO_RAW
LCD_MOSI
LCD_SCLK
LCD_CS
LCD_DC
LCD_RST
LCD_BL
TP_CS
LCD_TP_IRQ_RAW
TP1
PANEL_3V3
TP2
GND
TP3
SCL
TP4
SDA
TP5
IO_RESET
TP6
LCD_5V
R31
10k
IO_RESET
SW9
IO_RESET
GND
I/O RESET ONLY
C14
100n
C15
10u / 50V
H5
M3 LCD
H6
M3 LCD
H7
M3 LCD
H8
M3 LCD
H9
M2.5 PI STACK
H10
M2.5 PI STACK
H11
M2.5 PI STACK
H12
M2.5 PI STACK
H1
M3 PANEL
H2
M3 PANEL
H3
M3 PANEL
H4
M3 PANEL
R45
4.7k
SDA
R44
4.7k
SCL
D1
3V3 POWER
GND
LED1_A
R27
2.2k
LED1_A
D2
PROTO FAULT
BB_FAULT
LED2_A
R28
2.2k
LED2_A
R32
33
HAT_MISO
ISO_MISO
R33
33
ISO_MOSI
LCD_MOSI
R34
33
ISO_SCLK
LCD_SCLK
R35
100
ISO_LCD_CS
LCD_CS
R36
100
ISO_LCD_DC
LCD_DC
R37
100
ISO_LCD_RST
LCD_RST
R38
100
ISO_LCD_BL
LCD_BL
R39
100
ISO_TP_CS
TP_CS
R40
10k
LCD_VIO
LCD_CS
R41
10k
LCD_VIO
TP_CS
R42
100k
GND
LCD_BL
R43
22k
LCD_VIO
LCD_TP_IRQ_RAW
K1
AQV212EH / INPUT DISCONNECT
METER_RELAY_A
GND
METER_SWITCHED
METER_PROBE_PLUS
R52
49.9k / 0.1%
METER_SWITCHED
METER_RMID
R53
49.9k / 0.1%
METER_RMID
METER_DIV
R54
10.0k / 0.1%
METER_DIV
GND
R55
1k
METER_DIV
METER_AMP_IN
C21
100n
METER_DIV
GND
D3
BAT85
METER_AMP_IN
GND
D4
BAT85
PANEL_3V3
METER_AMP_IN
U3A
MCP6002-I/P
METER_AMP_IN
METER_BUFFER
METER_BUFFER
U3B
MCP6002-I/P
REF_2V5
REF_BUFFER
REF_BUFFER
R56
100
METER_BUFFER
ADC_CH0
C22
1n
ADC_CH0
GND
R57
100
REF_BUFFER
ADC_CH1
C23
1n
ADC_CH1
GND
U3C
MCP6002-I/P
PANEL_3V3
GND
U2
MCP3202-BI/P
PANEL_3V3
ADC_CH0
ADC_CH1
CH0
CH1
VDD/VREF
CS/SHDN
CLK
DOUT
DIN
VSS
ADC_CS
ADC_SCLK
ADC_MISO
ADC_MOSI
GND
C24
100n
GND
C25
100n
GND
R1
10k
KEY1
R9
1k
BTN1
C1
100n
SW1
GND
F1
R2
10k
KEY2
R10
1k
BTN2
C2
100n
SW2
GND
F2
R3
10k
KEY3
R11
1k
BTN3
C3
100n
SW3
GND
F3
R4
10k
KEY4
R12
1k
BTN4
C4
100n
SW4
GND
F4
R5
10k
KEY5
R13
1k
BTN5
C5
100n
SW5
GND
F5
R6
10k
KEY6
R14
1k
BTN6
C6
100n
SW6
GND
F6
R7
10k
KEY7
R15
1k
BTN7
C7
100n
SW7
GND
F7
R8
10k
HOLD_RAW
R16
1k
HOLD_IN
C8
100n
SW8
HOLD REQUEST
R46
10k
KEY8
R49
1k
BTN8
C18
100n
GND
R50
1k
BTN9
C19
100n
GND
R51
1k
BTN10
C20
100n
GND
SW10
KEY8
GND
SW11
KEY9
GND
SW12
KEY10
GND
U4
MCP1525-I/TO
PANEL_3V3
VIN
VOUT
REF_2V5
GND
C26
100n
GND
C27
4.7u / 50V
REF_2V5
GND
C28
100n
GND
R58
10k
ADC_CS
METER_LED_SUPPLY
R59
330 / 1%
METER_RELAY_A
PANEL_3V3
R60
10k
METER_EN_RAW
METER_EN_RAW
R61
1k
METER_EN_IN
PROTO_5V
R62
2.2k
PROTO_LED_A
ADC_MISO
R63
33
HAT_MISO
HAT_MOSI
R64
100
ADC_MOSI
HAT_SCLK
R65
100
ADC_SCLK
HAT_ADC_CS
R66
100
ADC_CS
SW13A
METER_LED_SUPPLY
PANEL_3V3
METER ENABLE
SW13B
METER_EN_RAW
GND
METER ENABLE
PROTO_5V
C29
100n
GND
PROTO_5V
GND
J13
PROTO 5V / GND
PROTO_5V
C30
10u / 50V
GND
J9
PROBE + / COMMON GND
GND
PROTO_LED_A
METER_PROBE_PLUS
GND
D5
PROTO POWER
SW14A
BB_EN
BB_EN
GND
PROTO 5V OFF / ON (100mA)
SW14B
PROTO 5V OFF / ON (100mA)
U5
TPS2553DRV
PI_5V_IN
BB_EN
BB_ILIM
IN
EN
ILIM
OUT
FAULT
PAD
PROTO_5V
BB_FAULT
GND
U6
Adafruit 4683 3V3 BUCK
GND
PANEL_3V3
PI_5V_IN
J11
LCD ACTUAL VIO / GND TAP
LCD_VIO
GND
J13
HAT J10 POWER / 1 GND 2 3V3
GND
PANEL_3V3
C31
100n
GND
PI_5V_IN
C32
100u / 16V
GND
BB_ILIM
R67
210k / 1%
GND
PI_5V_IN
R68
10k
BB_EN
BB_EN
R69
100k
GND
PANEL_3V3
R70
10k
BB_FAULT
PROTO_5V
R71
10k
GND
LCD_VIO
R72
1k
LCD_OE_N
LCD_VIO
R73
4.7k
GND
PANEL_3V3
R74
10k
HAT_LCD_CS
PANEL_3V3
R75
10k
HAT_TP_CS
HAT_MOSI
R76
100k
GND
HAT_SCLK
R77
100k
GND
HAT_LCD_DC
R78
100k
GND
HAT_LCD_RST
R79
100k
GND
HAT_LCD_BL
R80
100k
GND
LCD_MISO_RAW
R81
100k
GND
U8A
SN74LV1T34VR
HAT_MOSI
ISO_MOSI
LCD_OE_N
U8B
SN74LV1T34VR
HAT_SCLK
ISO_SCLK
LCD_OE_N
U8C
SN74LV1T34VR
HAT_TP_CS
ISO_TP_CS
LCD_OE_N
U8D
SN74LV1T34VR
HAT_LCD_DC
ISO_LCD_DC
LCD_OE_N
U8E
SN74LV1T34VR
LCD_VIO
VCC
GND
GND
C33
100n
LCD_VIO
GND
U9A
SN74LV1T34VR
HAT_LCD_RST
ISO_LCD_RST
LCD_OE_N
U9B
SN74LV1T34VR
HAT_LCD_BL
ISO_LCD_BL
LCD_OE_N
U9C
SN74LV1T34VR
GND
ISO_TP_CS
LCD_OE_N
U9D
SN74LV1T34VR
GND
LCD_VIO
U9E
SN74LV1T34VR
PANEL_3V3
VCC
GND
GND
C34
100n
LCD_VIO
GND
U10A
SN74LV1T34VR
LCD_MISO_RAW
ISO_MISO
HAT_TP_CS
U10B
SN74LV1T34VR
GND
LCD_OE_N
GND
U10C
SN74LV1T34VR
GND
PANEL_3V3
U10D
SN74LV1T34VR
GND
PANEL_3V3
U10E
SN74LV1T34VR
PANEL_3V3
VCC
GND
GND
C35
100n
GND
U11A
SN74AHC1G08
LCD_TP_IRQ_RAW
IRQ_INVERTED
U11B
SN74AHC1G08
IRQ_INVERTED
HAT_TP_IRQ
U11C
SN74AHC1G08
LCD_VIO
LCD_VIO_ABSENT
U11D
SN74AHC1G08
LCD_VIO_ABSENT
LCD_VIO_PRESENT
U11E
SN74AHC1G08
GND
U11F
SN74AHC1G08
GND
U11G
SN74AHC1G08
PANEL_3V3
VCC
GND
GND
C36
100n
GND
#PWR023
PWR_FLAG
PI_5V_IN
#PWR024
PWR_FLAG
LCD_VIO
J12
HAT J4.3 GPIO6 / J4.4 GPIO7
LCD_VIO_PRESENT
BB_FAULT

0–24 V DC only, shared GND. No independent motor stop.

Selzo
Sheet: /
File: szscreen.kicad_sch
Title: szscreen – reference console / revised power and isolation
Size: User | Date: 2026-10-29
KiCad E.D.A. 10.0.1
Rev: 23 ENGINEERING / NOT
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